

Hyphae detection across species × medium groups

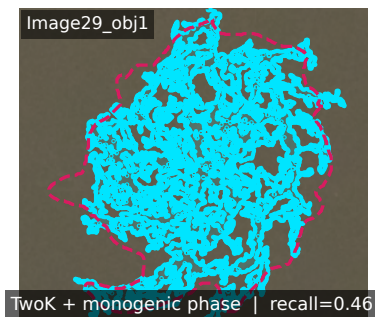
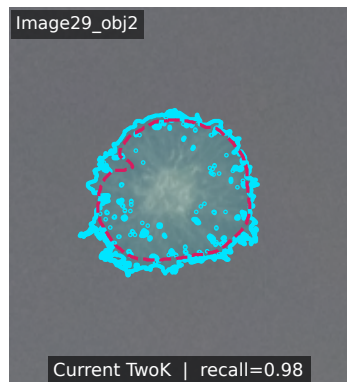
Qualitative human envelopes; candidate paths are species × medium prototypes

N. crassa
menadione

N. crassa
xylan

Ganoderma
glucose + yeast extract

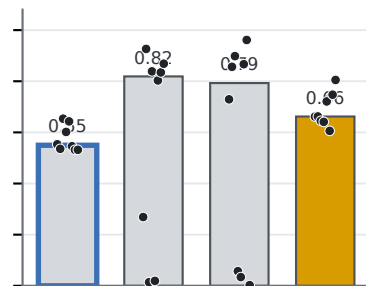
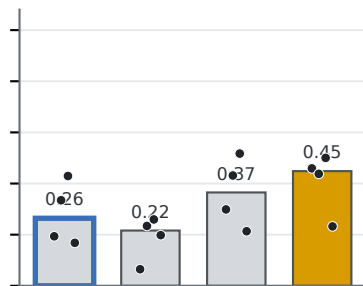
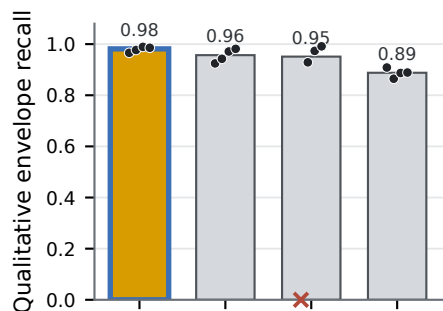
A



B

All-crop detector coverage

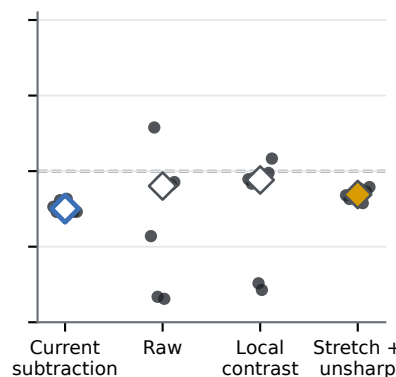
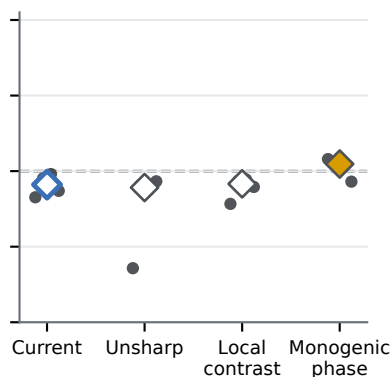
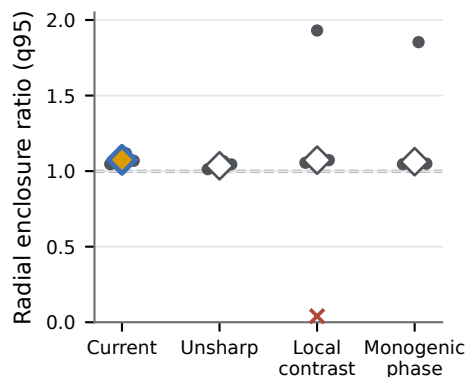
Gold = candidate; blue border = current baseline; dots = crops



C

Radial reach relative to the qualitative envelope

Diamond = median; dashed line = equal radial reach



Same-set evaluation: n=4 menadione, n=4 xylan, n=8 Ganoderma.
Human masks are subjective Sholl-style envelopes, not pixel ground truth.